

A Preserved-Extreme Tensor Need Not Be Preserved Extreme

Candidate counterexample to Question 3.9 of arXiv:2211.13559

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Status: candidate counterexample, likely valid. The construction gives a full negative answer to Question 3.9, subject to human review.

1 Source question

Garcia-Lirola, Grelier, Martinez-Cervantes, and Rueda Zoca ask the following on page 13 of [1].

Question 3.9. Is $x \otimes y$ a preserved extreme point of $B_{X \widehat{\otimes}_\pi Y}$ whenever x and y are preserved extreme points of B_X and B_Y ?

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isomorphic (it follows for instance from [19, Proposition 2.3]) and therefore $((-e_1 + e_n) \otimes (-e_1 + e_{k_n}))_n$ is not weakly convergent to $e_1 \otimes e_1$.

We end the paper with some open questions.

Question 3.9. Is $x \otimes y$ a preserved extreme point of $B_{X \widehat{\otimes}_\pi Y}$ whenever x and y are preserved extreme points of B_X and B_Y ?

Question 3.10. Is every (preserved) extreme point of $B_{X \widehat{\otimes}_\pi Y}$ a basic tensor?

REFERENCES

- [1] R. J. Aliaga, *Extreme points in Lipschitz-free spaces over compact metric spaces*, Mediterr. J. Math. **19** (2022), article 32.
- [2] R. J. Aliaga and A. J. Guirao, *On the preserved extremal structure of Lipschitz-free spaces*, J. Math. Anal. Appl. **475** (2019), 1–14.

Figure 1: Questions 3.9 and 3.10 in arXiv:2211.13559, PDF page 13.

All spaces below are real. Recall that $x \in B_X$ is *preserved extreme* if its canonical image is an extreme point of $B_{X^{**}}$.

2 Idea of the counterexample

Weak exposure and norm exposure have very different tensor behavior. We renorm ℓ_2 so that the points

$$(1 - \varepsilon_n)e_0 + e_n \quad \text{and} \quad (1 - \varepsilon_n)e_0 - e_n$$

belong to the unit ball, where $\varepsilon_n \downarrow 0$. They both converge weakly, but not in norm, to e_0 . The e_0 -coordinate nevertheless weakly strongly exposes e_0 : near-maximizers must place asymptotically all their mass on high-index generators, and those generators converge weakly to e_0 . This already implies that e_0 is preserved extreme.

Tensoring preserves the hidden norm-sized perturbations. Correlate the positive perturbation in the second factor once with the positive perturbation and once with the negative perturbation in the first factor. The two tensor sequences have distinct weak-star cluster points, detected by the Hilbert inner product, while their midpoint loses the perturbation and converges to $e_0 \otimes e_0$. Hence $e_0 \otimes e_0$ is not extreme in the tensor bidual.

3 The renorming

Let $H = \ell_2(\{0, 1, 2, \dots\})$ over $\mathbb{R}$, with orthonormal basis $(e_n)_{n \geq 0}$. For $n \geq 1$, put

$$\varepsilon_n = \frac{1}{n+1}, \quad p_n^\sigma = (1 - \varepsilon_n)e_0 + \sigma e_n \quad (\sigma \in \{-1, 1\}).$$

Define

$$G = \frac{1}{2}B_H \cup \{\tau p_n^\sigma : n \geq 1, \sigma, \tau \in \{-1, 1\}\}, \quad B = \overline{\text{co}}(G), \quad (1)$$

where closure is in the Hilbert norm. Let $\|\cdot\|_X$ be the Minkowski functional of B and write $X = (H, \|\cdot\|_X)$.

Lemma 1. *The set B is the closed unit ball of an equivalent norm on H , and*

$$\frac{1}{\sqrt{2}}\|z\|_H \leq \|z\|_X \leq 2\|z\|_H \quad (z \in H). \quad (2)$$

Moreover, $e_0 \in B$.

Proof. The set G is symmetric, and every p_n^σ has Hilbert norm at most $\sqrt{2}$. Hence

$$\frac{1}{2}B_H \subseteq B \subseteq \sqrt{2}B_H.$$

Thus B is the closed unit ball of an equivalent norm and (2) follows. Finally,

$$\frac{p_n^1 + p_n^{-1}}{2} = (1 - \varepsilon_n)e_0 \longrightarrow e_0$$

in Hilbert norm, so closedness gives $e_0 \in B$. □

Let $f \in H^*$ be the zeroth-coordinate functional, $f(z) = \langle z, e_0 \rangle_H$.

Lemma 2. *The functional f has $\|f\|_{X^*} = 1$, $f(e_0) = 1$, and every net $(z_\alpha) \subset B$ satisfying $f(z_\alpha) \rightarrow 1$ converges weakly to e_0 . Consequently, e_0 is a preserved extreme point of B .*

Proof. On $\frac{1}{2}B_H$ we have $f \leq \frac{1}{2}$, while

$$f(p_n^\sigma) = 1 - \varepsilon_n, \quad f(-p_n^\sigma) = -1 + \varepsilon_n.$$

Together with Lemma 1, this proves $\sup_B f = f(e_0) = 1$.

We prove the asserted weak convergence by a uniform slice estimate. Fix $h = (h_j) \in H$, viewed as the Hilbert functional $z \mapsto \langle z, h \rangle$, and fix $\eta > 0$. Choose N so large that

$$\varepsilon_n|h_0| + |h_n| < \eta \quad (n > N). \quad (3)$$

Call the generators p_n^σ with $n > N$ good and all other elements of G bad. Every good generator g satisfies

$$|\langle g - e_0, h \rangle| < \eta. \quad (4)$$

Every bad generator has a uniform deficit

$$1 - f(g) \geq c_N := \min\{1/2, \varepsilon_N\} > 0. \quad (5)$$

If $z = \sum_j \lambda_j g_j$ is a finite convex combination of generators and $f(z) > 1 - \delta$, (5) shows that the total coefficient of its bad generators is at most δ/c_N . Since

$$C_h := \sup_{g \in G} |\langle g - e_0, h \rangle| < \infty,$$

(4) gives

$$|\langle z - e_0, h \rangle| \leq \eta + \frac{\delta}{c_N} C_h. \quad (6)$$

Approximation by finite convex combinations extends (6), with an arbitrarily small error, to all $z \in B$. Letting first $\delta \downarrow 0$ and then $\eta \downarrow 0$ proves that $f(z_\alpha) \rightarrow 1$ forces $z_\alpha \rightarrow e_0$ weakly.

For completeness, suppose

$$e_0 = \frac{x_1^{**} + x_2^{**}}{2}, \quad x_1^{**}, x_2^{**} \in B_{X^{**}}.$$

Evaluation by f forces $f(x_i^{**}) = 1$. By Goldstine's theorem, approximate each x_i^{**} weak-star by a net in B . The preceding paragraph makes that net converge weakly to e_0 , and therefore its weak-star limit in X^{**} is e_0 . Thus $x_1^{**} = x_2^{**} = e_0$, proving preserved extremality. $\square$

4 Failure in the projective tensor bidual

Theorem 3. *For the equivalent renorming X above, e_0 is a preserved extreme point of B_X , but $e_0 \otimes e_0$ is not a preserved extreme point of $B_{X \widehat{\otimes}_\pi X}$. Therefore Question 3.9 has a negative answer.*

Proof. Set

$$r_n = p_n^1 = (1 - \varepsilon_n)e_0 + e_n, \quad s_n = p_n^{-1} = (1 - \varepsilon_n)e_0 - e_n,$$

and, in $Z = X \widehat{\otimes}_\pi X$, define

$$U_n = r_n \otimes r_n, \quad V_n = s_n \otimes r_n. \quad (7)$$

Both r_n and s_n belong to B_X , so $U_n, V_n \in B_Z$. By weak-star compactness of $B_{Z^{**}} \times B_{Z^{**}}$, there is a common subnet along which

$$U_n \xrightarrow{w^*} U, \quad V_n \xrightarrow{w^*} V \quad (U, V \in B_{Z^{**}}). \quad (8)$$

The midpoint has the exact form

$$\frac{U_n + V_n}{2} = \frac{r_n + s_n}{2} \otimes r_n = (1 - \varepsilon_n)e_0 \otimes ((1 - \varepsilon_n)e_0 + e_n). \quad (9)$$

The dual Z^* is the space of bounded bilinear forms on $X \times X$. For any such form A , evaluation of (9) is

$$(1 - \varepsilon_n)^2 A(e_0, e_0) + (1 - \varepsilon_n) A(e_0, e_n) \longrightarrow A(e_0, e_0),$$

because (e_n) is weakly null in X (the norm of X is equivalent to the Hilbert norm). Hence (9) converges weak-star to $e_0 \otimes e_0$. Passing to the subnet in (8) yields

$$e_0 \otimes e_0 = \frac{U + V}{2}. \quad (10)$$

It remains to show that this midpoint is nontrivial. The original Hilbert inner product

$$Q(a, b) = \langle a, b \rangle_H$$

is a bounded bilinear form on $X \times X$ by (2). From (7),

$$Q(U_n) = (1 - \varepsilon_n)^2 + 1 \longrightarrow 2, \quad Q(V_n) = (1 - \varepsilon_n)^2 - 1 \longrightarrow 0.$$

Therefore $Q(U) = 2$ and $Q(V) = 0$, so $U \neq V$. Equation (10) is a nontrivial midpoint decomposition inside $B_{Z^{**}}$. Thus $e_0 \otimes e_0$ is not preserved extreme. Lemma 2 verifies that both of its factors satisfy the hypothesis of Question 3.9. $\square$

5 Limitations and novelty check

The theorem completely answers Question 3.9 in the negative. It does not answer Question 3.10: the tensor in the counterexample is basic, and the argument concerns failure of preserved extremality rather than existence of a non-basic extreme tensor.

A bounded arXiv search on 22 July 2026 used the exact source-question phrase, the source arXiv id, and combinations of “preserved extreme”, “projective tensor”, “counterexample”, and “weak-strongly exposed”. It located the source and the later positive partial result [2], but no counterexample or full resolution. This supports, but does not certify, novelty.

6 Verification and human-review focus

The proof was checked at four sensitive points: equivalence of the new norm, the uniform weak slice estimate (6), simultaneous weak-star compactness in (8), and separation of the two limits by Q . No numerical computation is used. Human review should focus first on Lemma 2 and then on the midpoint convergence in (9).

References

- [1] L. C. Garcia-Lirola, G. Grelier, G. Martinez-Cervantes, and A. Rueda Zoca, *Extremal structure of projective tensor products*, Results Math. 78 (2023), article 196; arXiv:2211.13559.
- [2] R. J. Aliaga, L. C. Garcia-Lirola, J. Guerrero-Viu, M. Raja, and A. Rueda Zoca, *Almost preserved extreme points*, arXiv:2510.21234, 2025.